

SM015

**Past Year's
Question
Bank**

compiled by sir anam

Name:

Tutorial:

2012/13

1. Find the value of x which satisfies the equation $\log_2(5-x) - \log_2(x-2) = 3 - \log_2(1+x)$. [6 marks]
2. Determine the solution set of the inequality $\frac{1}{2x-1} < \frac{1}{x+2}$. [6 marks]
3. Given a complex number $z = 1 - \sqrt{3}i$. Determine the value of k if $\overline{z^2} = k \frac{1}{z}$. [7 marks]
4. (a) Solve $|x^2 - x - 3| = 3$. [5 marks]
- (b) Find the solution set of the inequality $\frac{2x^2 + 9x - 4}{x+2} < 4$. [7 marks]

5. Given $f(x) = \ln(2x+3)$ and $g(x) = \frac{e^x - 3}{2}$.
 - (a) Show that $f(x)$ is one-to-one algebraically. [3 marks]
 - (b) Find $(f \circ g)(x)$ and $(g \circ f)(x)$. Hence, state the conclusion about the result. [5 marks]
 - (c) Sketch the graphs of $f(x)$ and $g(x)$ on the same axes. Hence, state the domain and range of $f(x)$. [5 marks]
6. Given that

$$f(x) = \begin{cases} 1 + e^x & x < 1 \\ 1 & x = 1 \\ 2 - x & x > 1 \end{cases}$$

Find $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$. Does the $\lim_{x \rightarrow 1} f(x)$ exist? State your reason. [5 marks]

7. Find the following limits:

- (a) $\lim_{x \rightarrow \infty} \frac{2x^2 + x - 4}{1 - x^2}$ [3 marks]

- (b) $\lim_{x \rightarrow 2} \frac{3 - \sqrt{x+7}}{x^2 - 4}$ [4 marks]

8. (a) Given that $f(x) = \begin{cases} \frac{|x^2 - x - 2|}{x^2 - 2x} & x \neq 0, 2 \\ 0 & x = 2 \end{cases}$

Find the $\lim_{x \rightarrow 2} f(x)$. Is $f(x)$ continuous at $x = 2$? [6 marks]

(b) A function $f(x)$ is defined by $f(x) = \begin{cases} \alpha x + 6 & x < 4 \\ x^2 + 2 & 4 \leq x < 6 \\ 2 - \beta x & x \geq 6 \end{cases}$

Determine the values of the constants α and β if $f(x)$ is continuous. [5 marks]

9. (a) Given that $y = \frac{1}{\sqrt{2x+1}}$.

(i) By using the first principle of derivative, find $\frac{dy}{dx}$. [4 marks]

(ii) Find $\frac{d^2y}{dx^2}$. [2 marks]

(b) Find the following

(i) $y = e^{2x} \tan x$. [2 marks]

(ii) $y = x^{\sec x}$ [4 marks]

10. (a) A conical tank of height 12 m and surface diameter 8m. Water is pumped into the tank at the rate of $50 \text{ m}^3 / \text{min}$. How fast is the water level increasing when the depth of the water is 6 m ? [6 marks]

(b) A cylindrical container of radius r and height h has a constant volume V . The cost of the material for the surface of both of its ends is twice the cost of its sides. State h in terms of r and V . Hence, find h and r in terms of V such that the cost is minimum. [7 marks]

11. (a) Given $3y^2 - xy + x^2 = 3$. By using implicit differentiation,

(i) Find the value of $\frac{dy}{dx}$ at $x = 1$. [6 marks]

(ii) Show that $(6y - x)\frac{d^2y}{dx^2} + 6\left(\frac{dy}{dx}\right)^2 - 2\frac{dy}{dx} + 2 = 0$ [2 marks]

(c) Consider the parametric equations

$$x = 3t - \frac{2}{t}, y = 3t + \frac{2}{t} \text{ where } t \neq 0$$

(i) Show that $\frac{dy}{dx} = 1 - \frac{4}{3t^2 + 2}$ [3 marks]

(ii) Find $\frac{d^2y}{dx^2}$ when $t = 1$ [4 marks]

FINAL ANSWER:

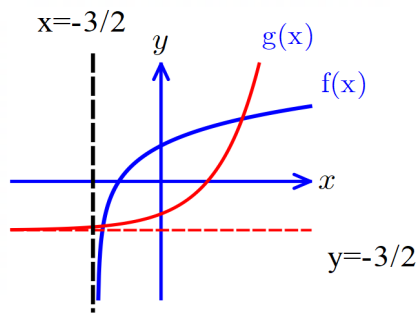
Q1 $x = 3$

Q2 $\left\{x : -2 < x < \frac{1}{2} \text{ or } x > 3\right\}$

Q3 $k = -8$

Q4 (a) $x = 3, -2 \text{ or } x = 0, 1$ (b) $\left\{x : x < -4 \text{ or } -2 < x < \frac{3}{2}\right\}$

Q5 (a) *DIY* (b) $f \circ g(x) = x$, $g \circ f(x) = x$, *conclusion DIY*
(c)



Q6 *DIY*

Q7 (a) -2 (b) $-\frac{1}{24}$

Q8 (a) *DIY* (b) $\alpha = 3$, $\beta = -6$

Q9 (a)(i) $-\frac{1}{(2x+1)^{\frac{3}{2}}}$ (ii) $\frac{3}{(2x+1)^{\frac{5}{2}}}$

(b)(i) $\frac{dy}{dx} = e^{2x}(\sec^2 x + 2 \tan x)$ (ii) $\frac{dy}{dx} = x^{\sec x} \sec x \left(\frac{1 + x \ln x \tan x}{x} \right)$

Q10 (a) $\frac{dh}{dt} = \frac{25}{2\pi}$

(b) $h = \frac{V}{\pi r^2}$, h and r in term of V such that the cost is min, $h = \left(\frac{16V}{\pi} \right)^{\frac{1}{3}}$

Q11 (a)(i) $\frac{dy}{dx} = \frac{8}{15}$ & $\frac{dy}{dx} = -\frac{1}{5}$ (ii) *DIY*

(b)(i) *DIY* (ii) $\frac{d^2y}{dx^2} = \frac{24}{125}$

2013/14

1. Consider the function $f(x) = 1 + \ln x, x \geq 1$. Determine $f^{-1}(x)$ and state its range. Hence evaluate $f^{-1}(3)$. [6 marks]
2. Find the value of x which satisfies the equation $\log_9 x = (\log_3 x)^2, x > 1$. [7 marks]
3. Solve the equation $2^{2x-2} - 2^{x+1} = 2^x - 2^3$. [7 marks]
4. Given $g(x) = \frac{kx + 8}{4x - 5}, x \neq \frac{5}{4}$ where k is a constant.
 - (a) Find the value of k if $(g \circ g)(x) = x$. [5 marks]
 - (b) Find the value of k so that $g(x)$ is not a one-to-one function. [5 marks]
5. Given $f(x) = e^{3x} + 4, x \in R$
 - (a) Find $f^{-1}(x)$. [5 marks]
 - (b) On the same axes, sketch the graphs of $f(x)$ and $f^{-1}(x)$. State the domain of $f(x)$ and $f^{-1}(x)$. [6 marks]
6. (a) Find the values of p and q if $\frac{p}{4-2i} + \frac{q}{4+2i} = 1 + \frac{5}{2}i$. [6 marks]
 - (b) Given $\log_{10} 2 = m$ and $\log_{10} 7 = n$. Express x in terms of m and n if $(14^{3x+1})(8^{2x+3}) = 7$. [6 marks]
7. (a) Find the solution set of $|2 - 3x| < |x + 3|$. [8 marks]
 - (b) If $x + 1 < 0$, show that
 - (i) $2x - 1 < 0$. [3 marks]
 - (ii) $\frac{2x-1}{x+1} > 2$. [4 marks]
8. (a) Find the value of m if $\lim_{x \rightarrow 0} \frac{mx + 3x^2}{4x - 8x^2} = 3$. [3 marks]
 - (b) Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{3-x} - \sqrt{3}}{x}$. [4 marks]
9. (a) Find $\frac{dy}{dx}$ if $y = \operatorname{cosec}\{\sin[\ln(x+1)]\}$. [3 marks]
 - (b) Obtain the second derivative of $y = \frac{\cos 3x}{e^{2x}}$ and express your answer in the simplest form. [4 marks]

10. (a) State the definition of the continuity of a function at a point. Hence, find the value of d such that

$$f(x) = \begin{cases} e^{3x+d} & x \leq 0 \\ 3x + 5 & x > 0 \end{cases}$$

is continuous at $x = 0$.

[5 marks]

- (b) A function f is define by

$$f(x) = \begin{cases} x^2 - 1 & x \leq 1 \\ k(x - 1) & x > 1 \end{cases}$$

Determine the value(s) of k if f is:

- (i) Continuous for all $x \in R$.

[3 marks]

- (ii) Differentiable for all $x \in R$.

[4 marks]

11. (a) Find the derivative of $f(x) = \frac{1}{x+1}$ by using the first principle. [4 marks]

- (b) Use implicit differentiation to find

- (i) $\frac{dy}{dx}$ if $y \ln x = e^{x-y}$. [3 marks]

- (ii) The value of $\frac{dy}{dx}$ if $\frac{1}{y} - \frac{1}{x} = 3$ when $x = \frac{1}{2}$. [5 marks]

12. A curve is defined by parametric equations

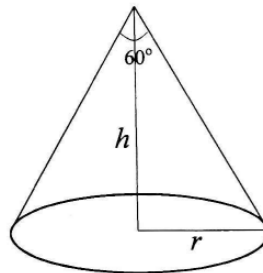
$$x = \ln(1+t), y = e^{x^2} \text{ for } t > -1.$$

- (a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of t . [6 marks]

- (b) Show that the curve has only one relative extremum at $(0,1)$ and determine the nature of the point. [6 marks]

13. (a) A cylindrical container of volume $128\pi m^3$ is to be constructed with the same material for the top, bottom and lateral side. Find the dimension of the container that will minimise the amount of the material needed. [6 marks]

- (c) Gravel is poured onto a flat ground at the rate of $\frac{3}{20} m^3$ per minute to form a conical-shaped with vertex angle 60° as shown in the diagram below



Compute the rate of change of the height of the conical pile at the instant $t = 10$ minutes. [7 marks]

FINAL ANSWER

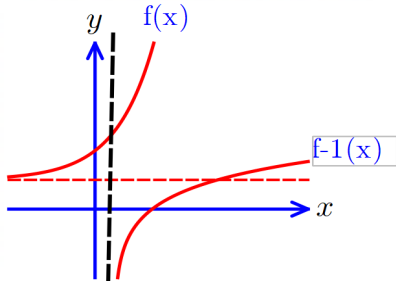
Q1 $f^{-1}(x) = e^{x-1}, f^{-1}(3) = e^2$

Q2 $x = \sqrt{3}$

Q3 $x = 3, x = 2$

Q4 (a) $k = 5$ (b) $k = -\frac{32}{5}$

Q5 (a) $f^{-1}(x) = \frac{1}{3} \ln(x-4)$



$D_f = \mathbb{R}, R_f = (-4, \infty)$

Q6 (a) $p = 15, q = -10$ (b) $x = \frac{-10m}{9m+3n}$

Q7 (a) $\left\{x : -\frac{1}{4} < x < \frac{5}{2}\right\}$ (b) (i) DIY (ii) DIY

Q8 (a) $m = 12$ (b) $-\frac{1}{2\sqrt{3}}$

Q9 (a) $\frac{dy}{dx} = \frac{-\operatorname{cosec}[\sin(\ln(x+1))]\cot[\sin(\ln(x+1))]\cos[\ln(x+1)]}{x+1}$

(b) $\frac{d^2y}{dx^2} = \frac{12\sin 3x - 5\cos 3x}{e^{2x}}$

Q10 (a) (i) $f(c)$ exist (ii) $\lim_{x \rightarrow c} f(x)$ exist (iii) $\lim_{x \rightarrow c} f(x) = f(c)$

$d = \ln 5$

(b) (i) $k \in R$ (ii) $k = 2$

Q11 (a) DIY (b) (i) $\frac{dy}{dx} = \frac{e^{x-y} - \frac{y}{x}}{\ln x + e^{x-y}}$ (ii) $\left.\frac{dy}{dx}\right|_{x=\frac{1}{2}} = \frac{4}{25}$

Q12 (a) $\frac{dy}{dx} = 2tet^2(1+t)$, $\frac{d^2y}{dx^2} = 2e^{t^2}(1+2t+2t^2+2t^3)[1+t]$

(b) $DIY, (0,1)$ is a relative minimum

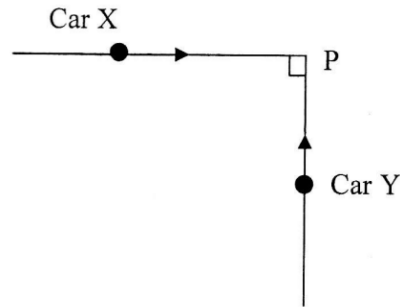
Q13 (a) $h = 8, r = 4$

(b) $\frac{dh}{dt} = 0.00542$

2014/15

1. Solve the equation $3^x + 3^{(3-x)} = 12$. [6 marks]
2. Solve the inequality $\frac{1}{6-x} < \frac{1}{x-1}$. [6 marks]
3. (a) Given $f(x) = 2x + 1$ and $g(x) = x^2 + 2x - 1$.
 - (i) Find $(f - g)(x)$. [2 marks]
 - (ii) Evaluate $(3g - 2f)(1)$. [4 marks]
- (b) Given $f(x) = \sqrt{2x + \frac{1}{2}}$. State the domain and range of $f(x)$. Hence, on the same axes, sketch the graph of $f(x)$ and $f^{-1}(x)$. [6 marks]
4. Let $z = a + bi$ be a nonzero complex number.
 - (a) Show that $\frac{1}{z} = \frac{\bar{z}}{|z|^2}$. [4 marks]
 - (b) Show that if $\bar{z} = -z$, then z is a complex number with only an imaginary part. [3 marks]
 - (c) Find the value of a and b if $z(2 - i) = (\bar{z} + 1)(1 + i)$. [5 marks]
5. (a) Solve the following equation $|6x^2 + x - 11| = 4$. [6 marks]
- (b) Find the solution set for the inequality $2 - \left(\frac{x+2}{x-4}\right) < 5$. [7 marks]
6. (a) Determine whether $f(x) = \frac{1}{x-4}$ and $g(x) = \frac{4x+1}{x}$ are inverse function of each other by computing their composite functions. [5 marks]
- (b) Given $f(x) = \ln(1 - 3x)$
 - (i) Determine the domain and range of $f(x)$. Then sketch the graph of $f(x)$. [6 marks]
 - (iii) Find $f^{-1}(x)$, if it exists. Hence, state the domain and range of $f^{-1}(x)$. [4 marks]
7. Find the relative extremum of the curve $y = x^3 - 4x^2 + 4x$. [6 marks]

8. Car X is travelling east at a speed of 80km/h and car Y is travelling north at 100km/h as shown in the diagram below. Obtain an equation that describes the rate of change of the distance between the two cars. Hence evaluate the rate of change of the distance between the two cars when car X is 0.15km and car Y is 0.08km from P .



[7 Marks]

9. Find the limit of the following if it exists.

(a) $\lim_{x \rightarrow 3} \frac{x+3}{x^3+27}$. [3 marks]

(b) $\lim_{x \rightarrow -\infty} \frac{2x-1}{\sqrt{x^2-9}}$. [3 marks]

(c) $\lim_{x \rightarrow 4} \frac{x^2-3x-4}{\sqrt{x}-2}$. [4 marks]

10. Given that

$$f(x) = \begin{cases} 1+e^x, & x \leq 0 \\ \frac{x+6}{3-x}, & 0 < x \leq 4 \\ C, & x > 4 \end{cases}$$

Where C is a constant.

- (a) Determine whether $f(x)$ is continuous at $x = 0$. [5 marks]
 (b) Given that $f(x)$ is discontinuous at $x = 4$, determine the values of C . [3 marks]
 (c) Find the vertical asymptote of $f(x)$. [4 marks]

11. Consider the parametric equation of the curve

$$x = \cos^3 \theta \text{ and } y = \sin^2 \theta, \quad 0 < \theta \leq 2\pi$$

- (a) Find $\frac{dy}{dx}$ and express your answer in terms of θ . [4 marks]
 (b) Find the value of $\frac{dy}{dx}$ if $x = \frac{\sqrt{2}}{4}$. [4 marks]
 (c) Show that $\frac{d^2y}{dx^2} = \frac{1}{3 \cos^4 \theta \sin \theta}$. Hence calculate $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{3}$. [5 marks]

12. (a) Use the first principle to find the derivative of $g(x) = \sqrt{1-x}$. [5 marks]
- (b) Given that $e^y + xy + \ln(1+2x) = 1, x \geq 0$. Show that
- $$(e^y + x) \frac{d^2y}{dx^2} + e^y \left(\frac{dy}{dx} \right)^2 + 2 \frac{dy}{dx} - \frac{4}{(1+2x)^2} = 0.$$
- Hence, find the value of $\frac{d^2y}{dx^2}$ at the point $(0,0)$. [10 marks]

FINAL ANSWER

Q1 $x = 2, x = 1$

Q7 $\left(\frac{2}{3}, \frac{32}{27} \right)$ is the relative maximum ,

$(2,0)$ is the relative minimum

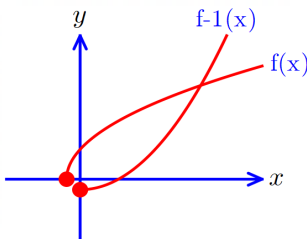
Q2 $\left(1, \frac{7}{2} \right) \cup (6, \infty)$

Q8 $\frac{dz}{dt} = \frac{1}{z} \left(x \frac{dy}{dx} + y \frac{dy}{dt} \right)$, $\frac{dz}{dt} = -117.647$

Q3 (a)(i) $-x^2 + 2$ (ii) $3x^2 + 2x - 5, 0$

Q9 (a) $\frac{1}{27}$ (b) -2 (c) 20

(b) $D_f = \left[-\frac{1}{4}, \infty \right)$ $R_f = [0, \infty)$



Q4 (a) *DIY* (b) *DIY* (c) $a = 1, b = 1$

Q10 (a) *DIY* (b) $c \neq -10$ (c) $x = 3$

Q5 (a) $x = -\frac{5}{3}, x = \frac{3}{2}, x = -\frac{7}{6}, x = 1$

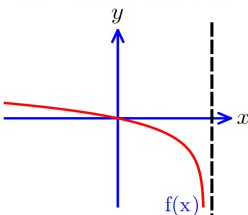
Q11 (a) $\frac{dy}{dx} = -\tan \theta$ (b) $\frac{dy}{dx} = 1$ (c) *DIY*, $\frac{d^2y}{dx^2} = 6.16$

(b) $\left\{ x : x < \frac{5}{2} \text{ or } x > 4 \right\}$

Q6 (a) *DIY*

Q12 (a) *DIY* (b) $\frac{d^2y}{dx^2} = 4$

(b)(i) $D_f = \left\{ x : x < \frac{1}{3} \right\}$ $R_f = \mathbb{R}$



(b)(ii) $f^{-1}(x) = \frac{1-e^x}{3}$

$D_{f^{-1}} = \mathbb{R}, R_{f^{-1}} = \left\{ y : y < \frac{1}{3} \right\}$

2015/16

1. Evaluate the solution of $4^{y-2} = \frac{1}{3^{-y}}$ up to three decimal places. [6 marks]
2. Solve the equation $2 + \log_2 x = 15 \log_x 2$. [7 marks]
3. Given $z_1 = 3 - 3i$ and $z_2 = 3 + 2i$.
 - (a) Write $\overline{z_1}$ in polar form. [4 marks]
 - (b) Express $\frac{\overline{(z_1 z_2)}}{13} + \left(\frac{i^3}{-z_2} \right)$ in the form of $a + bi$, $a, b \in R$. [8 marks]
4. Given a function $f(x) = \sqrt{3 - 2x}$
 - (a) Show that f is a one to one function. [2 marks]
 - (b) Find the domain and range of f . [3 marks]
 - (c) Determine the inverse function of f and state its domain and range. [4 marks]
 - (d) Sketch the graph of f and f^{-1} on the same axis. [3 marks]
5. (a) The function f is given as $f(x) = \frac{ax + 2}{3x - 4}$, $x \neq \frac{4}{3}$. If $(f \circ f)(x) = x$, find the value of a . [6 marks]
- (c) Let $f(x) = \ln|3x + 2|$ and $g(x) = e^{-x} + 2$ be two functions. Evaluate $(g \circ f)^{-1}(3)$. [7 marks]
6. (a) Solve the inequality $\left| \frac{x - 1}{x + 3} \right| > 2$. [6 marks]
- (b) Show that $\frac{2^x \times 4^{2x}}{8^x} = 2^{2x}$. Hence find the interval of x so that $\frac{2^x \times 4^{2x}}{8^x} - 13(2^x) + 36 \geq 0$. [9 marks]
7. Evaluate the following (if exist):
 - (a) $\lim_{x \rightarrow 2} \frac{x^2 + 4x - 12}{|x - 2|}$ [4 marks]
 - (b) $\lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{1 - x}$ [3 marks]
8. Find the derivative of the following functions:
 - (a) $f(x) = \cot \sqrt{4x^2 + 1}$ [3 marks]
 - (b) $f(x) = e^{2x} \ln(3x + 4)$ [4 marks]

9. (a) Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2 + 3x}}{5x + 1}$ [4 marks]

(b) Given $f(x) = \begin{cases} 5 - px & -2 < x \leq -1 \\ x^2 + px + q & -1 < x \leq 2 \\ \frac{x^2 - 4}{x - 2} & x > 2 \end{cases}$

(i) Find the values of p and q if $f(x)$ is continuous for all real values of x . [6 marks]

(ii) Sketch the graph of $f(x)$ using the values of p and q obtained in part (i) [3 marks]

10. A curve is given by the parametric equation

$$x = t - \frac{1}{t}, \quad y = t + \frac{1}{t}$$

(a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of t . [7 marks]

(b) Obtain the coordinate of the stationary points of the curve and determine the nature of the points. [6 marks]

11. (a) If $y^2 - 2y\sqrt{1+x^2} + x^2 = 0$, show that $\frac{dy}{dx} = \frac{x}{\sqrt{1+x^2}}$. [6 marks]

(b) Water is running at a steady rate of $36\pi m^3 s^{-1}$ into a right inverted circular cone with a semi-vertical angle of 45°

(i) Find the rate of increasing in water depth when the water level is $3cm$ [4 marks]

(ii) Find the time taken when the depth of the water is $18cm$. [3 marks]

12. (a) Find the value of k if the slope of the curve $x^3 + kx^2y - 2y^2 = 0$ at the point $(-1, 1)$ is -3 . [5 marks]

(b) Given $y = \frac{\sin x}{1 + \cos x}$

(i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of x . [5 marks]

(ii) Hence, show that $\frac{d^3y}{dx^3} - y \frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 = 0$. [5 marks]

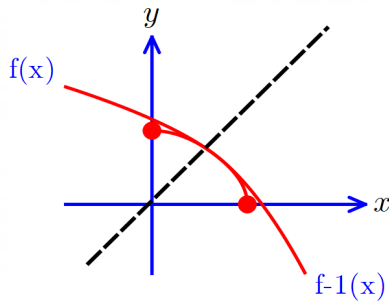
FINAL ANSWER

Q1 9.638

Q2 $x = 8, x = \frac{1}{32}$

Q3 (a) $3\sqrt{2}[\cos 0.785 + i \sin 0.785]$ (b) $\frac{5}{13} + \frac{12}{13}i$

Q4 (a) DIY (b) $D_f = \left[-\infty, \frac{3}{2}\right]$ $R_f = [0, \infty)$
 (c) $f^{-1}(x) = \frac{3-x^2}{2}$ $D_{f^{-1}} = [0, \infty)$ $R_{f^{-1}} = \left[-\infty, \frac{3}{2}\right]$



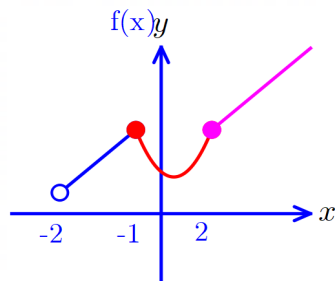
Q5 (a) $a = 4$ (b) $(g \circ f)^{-1}(x) = \frac{5-2x}{3x-6}, (g \circ f)^{-1}(3) = -\frac{1}{3}$

Q6 (a) $(-7, -3) \cup \left[-3, -\frac{5}{3}\right]$ (b) $(-\infty, 2] \cup [3.17, \infty)$

Q7 (a) does not exist (b) $\frac{1}{2}$

Q8 (a) $-\frac{4 \cos e c^2 \sqrt{(4x^2 + 1)}}{\sqrt{(4x^2 + 1)}}$ (b) $e^{2x} \left[\frac{3}{3x+4} + 2 \ln(3x+4) \right]$

Q9 (a) $-\frac{\sqrt{2}}{5}$ (b) (i) $p = -1, q = 2$



Q10 (a) $\frac{dy}{dx} = \frac{t^2 - 1}{t^2 + 1}, \frac{d^2y}{dx^2} = \frac{4t^3}{(t^2 + 1)^3}$ (b) $(0, -2)$ is a max. point $(0, 2)$ is a min. point

Q11 (a) DIY (b) (i) 4 cm s^{-1} (ii) $t = 54 \text{ s}$

Q12 (a) $k = 3$ (b) (i) $\frac{dy}{dx} = \frac{1}{1 + \cos x}, \frac{d^2y}{dx^2} = \frac{\sin x}{(1 + \cos x)^2}, (ii) DIY$

2016/17

1. Solve for p and q where $p \neq q$, such that $\frac{(p+qi)}{3i} = (3 + \sqrt{-16} - i^3)$. [6 marks]
2. Determine the values of x which satisfy the equation $3^{2x-1} = 4(3^x) - 9$. [6 marks]
3. (a) Given $f(x) = \frac{x^2+1}{5}, x \geq 0$. Determine $f^{-1}(x)$. Hence, if $f(g(x)) = \frac{1}{5}(e^{2(3x-1)} + 1)$, show that $g(x) = e^{3x-1}$. [5 marks]
- (b) Evaluate $g(f(2))$ correct to three decimal places. [3 marks]
- (c) Assume that the domain for $g(x)$ is $x \geq 0$, determine $g^{-1}(x)$ and state its domain and range. [3 marks]
4. (a) If $\sqrt{7-3\sqrt{5}} = \sqrt{x} - \sqrt{y}$. Determine the values of x and y . [6 marks]
- (b) Solve the equation $\log_2 x - \log_4(3x+4) = 0$. [7 marks]
5. (a) Solve the following equation $\left| \frac{3}{x-4} \right| = 7, x \neq 4$. [5 marks]
- (b) Find the solution set for the inequality $\frac{-4-x}{x-3} \geq x+4, x \neq 3$. [8 marks]
6. Given $f(x) = e^x$ and $g(x) = |x-3|$
 - (a) Show that $(f \circ g)(x) = \begin{cases} e^{x-3} & x \geq 3 \\ e^{-(x-3)} & x < 3 \end{cases}$ [2 marks]
 - (b) Sketch the graph of $y = (f \circ g)(x)$. Hence, state the interval which $(f \circ g)^{-1}(x)$ exists. [3 marks]
 - (c) Determine $(f \circ g)^{-1}(x)$ for $x \geq 3$ [3 marks]
 - (d) Find the function $h(x)$ for $x > \frac{1}{3}$, given that $(h \circ f)(x) = \frac{2e^x}{1-3e^x}$. Hence, show that $h(x)$ is a one-to-one function. [5 marks]
7. Evaluate the following limits, if exist.
 - (a) $\lim_{x \rightarrow 2} \frac{x-2}{x^4-16}$ [3 marks]

(b) $\lim_{x \rightarrow \infty} \frac{(2-x)(x-1)}{(x-3)^2}$ [3 marks]

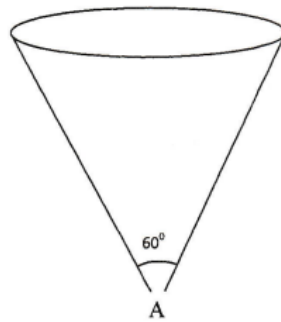
8. Consider a function $f(x) = \frac{1}{2 - \sqrt{x}}$

(a) Find $\lim_{x \rightarrow \infty} f(x)$ and state the equation of horizontal asymptote for f . [3 marks]

(b) By using the first principle of derivative, find $f'(x)$. [4 marks]

9. (a) Use the derivative to find the maximum area of a rectangle that can be inscribed in a semicircle of radius 10cm . [5 marks]

(b) A cone-shaped tank as shown below



Water flows through a hole A at a rate of 6cm^3 per second. Find the rate of change on height of the water when the volume of water in the cone is $24\pi\text{cm}^3$. [6 marks]

10. The parametric equations of a curve is given by

$$x = e^{2t+1}, y = e^{-(2t-1)}$$

(a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $t = 1$. [7 marks]

(b) Given $z = x^2 - xy$. Express z in terms of t and find $\frac{dz}{dt}$. Hence, deduce the set of value of t such that $\frac{dz}{dt}$ is positive. [5 marks]

11. (a) Given $f(x) = \frac{2|x|}{x} + 5x$. Compute $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$. Is f continuous at $x = 0$? Give your reason. [5 marks]

(b) The continuous function g is defined by $g(x) = \begin{cases} \sqrt{5-x} & x < a \\ 3x-1 & x \geq a \end{cases}$. Find the value of a . [8 marks]

12. By writing $\tan x$ in terms of $\sin x$ and $\cos x$, show that $\frac{d}{dx}(\tan x) = \sec^2 x$. [3 marks]

(a) If $y = \tan x$, find $\frac{d^2 y}{dx^2}$ in terms of y . Hence, determine the range of value of x such that $\frac{d^2 y}{dx^2} > 0$ for $0 < x < \pi$. [7 marks]

(b) If $y = \tan(x + y)$, find $\frac{dy}{dx}$ in terms of x and y . Hence, show that $\frac{dy}{dx} = \operatorname{cosec}^2 2\alpha$ when $x = y = \alpha$. [5 marks]

FINAL ANSWER

Q1 $p = -15, q = 9$

Q2 $x = 1, x = 2$

Q3 (a) $f^{-1}(x) = \sqrt{5x-1}$, DIY (b) $g(f(2)) = e^2$ (c) $g^{-1}(x) = \frac{\ln x + 1}{3}$

Q4 $D_{g^{-1}} = x \geq \frac{1}{e}$, $R_{g^{-1}} = y \geq 0$

(a) $x = \frac{9}{2}, y = \frac{5}{2}$ $x = \frac{5}{2}, y = \frac{9}{2}$
(b) $x = 4$

Q5 (a) $x = \frac{31}{7}, x = \frac{25}{7}$ (b) $\{x : -\infty < x \leq -4 \text{ or } 2 \leq x < 3\}$

Q6 (a) DIY (b) $(f \circ g)^{-1}(x)$ only can exist if $x \geq 3$ or $x < 3$
(c) $(f \circ g)^{-1}(x) = \ln x + 3$
(d) $h(x) = \frac{2x}{1-3x}$

Q7 (a) $\frac{1}{32}$ (b) -1

Q8 (a) $y = 0$ (b) $f'(x) = \frac{1}{2\sqrt{x}(2-\sqrt{x})^2}$

Q9 (a) $\text{Max. Area} = 100 \text{ cm}^2$ (b) $\frac{dh}{dt} = -\frac{1}{2\pi} \text{ cm s}^{-1}$

Q10 (a) $\frac{dy}{dx} = -\frac{1}{e^4}, \frac{d^2 y}{dx^2} = \frac{2}{e^7}$ (b) $z = e^{4t+2} - e^2, \frac{dz}{dt} = 4e^{4t+2}, \{t : t \in \mathbb{R}\}$

Q11 (a) DIY (b) $a = 1$

Q12 (a) $\frac{d^2 y}{dx^2} = 2y(1+y^2), \left(0, \frac{\pi}{2}\right)$ (b) $\frac{dy}{dx} = \frac{\sec^2(x+y)}{1-\sec^2(x+y)}$

2017/18

1. Solve the equation $3^{2x+1} - (16)3^x + 5 = 0$ [6 marks]
2. Solve the equation $3 \log_9 x = (\log_3 x)^2$. [9 marks]
3. Given a complex number $z = 2 + i$.
 - (a) Express $\bar{z} - \frac{1}{z}$ in the form of $a + bi$, where a and b are real numbers. [4 marks]
 - (b) Obtain $\left| \bar{z} - \frac{1}{z} \right|$. Hence, determine the values of real numbers α and β if

$$\alpha + \beta i = \left| \bar{z} - \frac{1}{z} \right| \left(\bar{z} - \frac{1}{z} \right)^2.$$
 [8 marks]
4. Find the interval of x for which the following inequalities are true.
 - (a) $\frac{5}{x+3} - 1 \leq 0$ [4 marks]
 - (b) $\left| \frac{3x-2}{2x+3} \right| > 2$ [9 marks]
5. Consider function of $f(x) = (x-2)^2 + 1, x > 2$ and $g(x) = \ln(x+1), x > 0$
 - (a) Find $f^{-1}(x)$ and $g^{-1}(x)$, and state the domain and range for each of the inverse function. [9 marks]
 - (b) Obtain $(g \circ f)(x)$. Hence, evaluate $(g \circ f)(2)$. [4 marks]
6. Given the function $g(x) = \frac{1}{2x-5}$
 - (a) Find the domain and range of $g(x)$. [3 marks]
 - (b) Show that $g(x)$ is a one-to-one function. Hence, find $g^{-1}(x)$. [5 marks]
 - (c) On the same axis, sketch the graph of $g(x)$ and $g^{-1}(x)$. [3 marks]
 - (d) Show that $g \circ g^{-1}(x) = x$. [2 marks]
7. Evaluate the following limits:
 - (a) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x}$ [3 marks]
 - (b) $\lim_{x \rightarrow \infty} \sqrt{\frac{5x+7}{6x-5}}$ [3 marks]

8. Given $y = e^{-2x} \sin 3x$. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. Hence, show that $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = 0$ [6 marks]
9. State the condition of continuity of $f(x)$ at $x = a$. [2 marks]
- (a) By using the conditions for continuity of $f(x)$ at $x = a$, find the values of m and n such that
- $$f(x) = \begin{cases} n - 2 \cos x & x < 0 \\ 2 + mx^2 & 0 \leq x < 2 \\ m - x & x \geq 2 \end{cases}$$
- Is continuous on the interval $(-\infty, \infty)$ [6 marks]
- (b) If $m = -2$ and $n = 4$, determine whether $f(x)$ is differentiable at $x = 2$ or not. [4 marks]
10. A curve with equation $x^2 - 3y^2 = ae^{y-2x} + by - 6$ where a and b are constants, passes through the point $(1, 2)$.
- (a) Given $\frac{dy}{dx} = 1$ at $(1, 2)$. Determine the values a and b . [8 marks]
- (b) Evaluate $\frac{d^2y}{dx^2}$ at $(1, 2)$ [4 marks]
11. The function f is defined by $f(x) = \frac{\ln(x-1)}{x-1}$ for $x > 1$
- (a) By considering the first and second derivative of $f(x)$, show that there is only one maximum point on the graph $y = f(x)$. [8 marks]
- (b) Use the result obtained in part (a) to state the exact coordinate of the maximum point. [2 marks]
- (c) Find the x coordinate of the function f when $\frac{d^2y}{dx^2} = 0$ [2 marks]

12. A curve is defined by the parametric equations $x = 3t - \frac{1}{t}$ and $y = t + \frac{3}{t}$ where $t \neq 0$

(a) Show that $\frac{dy}{dx} = \frac{t^2 - 3}{3t^2 + 1}$. Hence, find $\frac{d^2y}{dx^2}$. [8 marks]

(b) Show that $\frac{dy}{dx}$ can be expressed as $\frac{dy}{dx} = \frac{1}{3} - \frac{10}{3(3t^2 + 1)}$.

Hence, deduce that $-3 < \frac{dy}{dx} < \frac{1}{3}$ [7 marks]

FINAL ANSWER

Q1 $x = -1, x = 1.465$

Q2 $x = 1, x = 5.196$

Q3 (a) $\frac{8}{5} - \frac{6}{5}i$ (b) $\alpha = \frac{56}{25}$ $\beta = -\frac{192}{25}$

Q4 (a) $\{x : x < -3 \text{ or } x \geq 2\}$
 (b) $\left(-8, -\frac{3}{2}\right) \cup \left(-\frac{3}{2}, -\frac{4}{7}\right)$

Q5 (a) $f^{-1}(x) = \sqrt{x-1} + 2$
 $g^{-1}(x) = e^x - 1$
 $D_{f^{-1}} = (1, \infty)$ $R_{f^{-1}} = (2, \infty)$
 $D_{g^{-1}} = (0, \infty)$ $R_{g^{-1}} = (0, \infty)$
 (b) $(g \circ f)(x) = \ln[(x-2)^2 + 2]$
 $(g \circ f)(2) = \ln 2$

Q6 (a) $D_g = \left(-\infty, \frac{5}{2}\right) \cup \left(\frac{5}{2}, \infty\right)$
 $R_g = (-\infty, 0) \cup (0, \infty)$

Q7 (a) 6 (b) $\sqrt{\frac{5}{6}}$

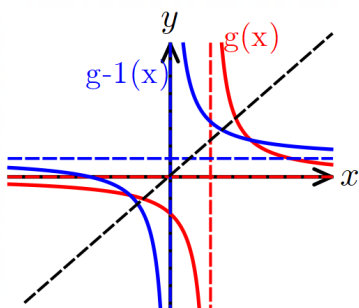
Q8 $\frac{dy}{dx} = e^{-2x} (3 \cos 3x - 2 \sin 3x)$
 $\frac{d^2y}{dx^2} = e^{-2x} [-5 \sin 3x - 12 \cos 3x]$

Q9 *DIY*
 (a) $m = -\frac{4}{3}, n = 4$

(b) $f(x)$ is differentiable at $x = 2$
 Q10 (a) $a = 5, b = -5$ (b) $\left. \frac{d^2y}{dx^2} \right|_{x=1} = -\frac{3}{4}$

Q11 (a) *DIY* (b) $\left(e + 1, \frac{1}{e}\right)$ (c) $x = e^{\frac{3}{2}} + 1$

Q12 (a) $\frac{d^2y}{dx^2} = \frac{20t^3}{(3t^2 + 1)^3}$ (b) *DIY*



2018/19

1. (a) Solve $\sqrt{6x+1} - \sqrt{x} = 3$ [6 marks]
- (b) Determine the solution set of x which satisfies the inequality $\frac{2}{x+1} < \frac{x}{x+3}$. [7 marks]
2. The parametric equations of a curve are $x = t + \frac{2}{t}$ and $y = 2t - \frac{4}{t}$, where $t \neq 0$. Show that $\frac{dy}{dx} = 2 + \frac{8}{t^2 - 2}$. Hence, find $\frac{d^2y}{dx^2}$ in terms of t . [7 marks]
3. Water is poured into a right inverted cone of height h with a semi-vertical angle of 60° at a constant rate of $25\pi\text{cm}^3$ per second.
 - (a) Show that the rate of change of the height of water is $\frac{dh}{dt} = \frac{25}{3h^2}$. [5 marks]
 - (b) Find the rate of change of the height of water after 5 seconds. [4 marks]
 - (c) Given the height of the cone is 20cm , find the time taken to fill the cone completely with water. [2 marks]
4. Given $z_1 = 2 + 3i$ and $z_2 = 4 - 4i$. Express $\frac{(z_2)}{(z_1)} + \left[\left(\frac{i^3}{-z_2} \right) \right]$. [8 marks]
5. Solve
 - (a) $\left(\frac{27}{125} \right)^2 \times \left(\frac{25}{9} \right)^{4x} = \left(\frac{9}{25} \right)^{x-3} \times \left(\frac{625}{81} \right)^2$ [6 marks]
 - (b) $\frac{1}{4-2x} \geq \frac{8}{x}$ [6 marks]
6. (a) Given $f(x) = \left(\frac{5x+1}{4x} \right)$ and $g(x) = \frac{\sqrt{x+1}-2}{x^2-4}$. Find
 - (i) the domain of $g(x)$
 - (ii) $h(x)$ if $(f \circ h)(x) = x$. [8 marks]
- (b) Given $p(x) = \ln(3x+6)$ and $q(x) = \frac{e^x}{3} - 2$. Show that $p(x)$ and $q(x)$ are inverse of each other. [6 marks]

7. The function $g(x)$ is defined by

$$g(x) = \begin{cases} 2 & x \leq 2 \\ \frac{x-2}{\sqrt{2x}-2} & 2 < x \leq 8 \\ \frac{|8-x|}{x-8} & x > 8 \end{cases}$$

Find

- (a) $\lim_{x \rightarrow 2^+} g(x)$ [5 marks]
- (b) $\lim_{x \rightarrow 8^+} g(x)$ [3 marks]
8. (a) Find the derivative of $f(x) = \frac{6}{\sqrt{x}}$ using the first principle. [5 marks]
- (b) Find the value of $\frac{dy}{dx}$ when $x = 0$ for each of the following
- (i) $y = \ln(9 - 2x)$ [3 marks]
- (ii) $y = \frac{e^{-3x}}{\sqrt{3x+1}}$ [3 marks]
9. Given $f(x) = \frac{3x}{x^2+9}$, where $x > 0$. Find the coordinates of the stationary point and state its nature. [5 marks]
10. Find the gradient of the curve $\cos(4xy) = \tan(xy^2) - 3y$ at the point where $x = 0$ [6 marks]

FINAL ANSWER

Q1 $(a) x = 4 \quad (b) \{x : x < -3 \text{ or } -2 < x < -1 \text{ or } x > 3\}$

Q2 $DIY, \frac{d^2y}{dx^2} = \frac{16t^3}{(t^2 - 2)^3}$

Q3 $(a) DIY \quad (b) \frac{dh}{dt} = \frac{1}{3} \text{ cm s}^{-1} \quad (c) t = 320 \text{ s}$

Q4 $\frac{147}{104} + \frac{45}{104}i$

Q5 $(a) x = 2 \quad (b) \left\{x : x < 0 \text{ or } \frac{32}{17} \leq x < 2\right\}$

Q6 $(a)(i) D_g = [-1, 2) \cup (2, \infty) \quad (ii) h(x) = \frac{-1}{5 - 4x}$

$(b) DIY$

Q7 $(a) 2 \quad (b) 1$

Q8 $(a) -\frac{3}{x^{\frac{3}{2}}} \quad (b)(i) -\frac{2}{9}, (ii) -\frac{9}{2}$

Q9 $\left(3, \frac{1}{2}\right)$ is a maximum point

Q10 $\frac{dy}{dx} = \frac{4y \sin(4xy) + y^2 \sec^2(xy^2)}{3 - 4x \sin(4xy) - 2xy \sec^2(xy^2)}$

$\frac{dy}{dx} \Big|_{x=0} = \frac{1}{27}$

2019/20

1. Given the complex numbers $z_1 = -i$ and $z_2 = 2 + i\sqrt{3}$
 - (a) Express $(z_1)^2$ and $\overline{z_2}$ in the form $a + bi$, where $a, b \in \mathbb{R}$ [2 marks]
 - (b) From part (a), find $W = \frac{(z_1)^2 + \overline{z_2}}{z_1}$. Hence, find $|W|$ and $\arg W$. [7 marks]
2. Solve the following
 - (a) $3(5^{2x}) + 25^{\frac{1}{2}x+1} = 200$ [5 marks]
 - (b) $x + 4 \leq x^2 + x < 12$ [5 marks]
3. Given $f(x) = \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}}$
 - (a) Simplify $f(x)$ and evaluate $f\left(\frac{1}{2}\right)$. [4 marks]
 - (b) The domain of $f(x)$ is a set of real number except three numbers. Determine the numbers. [4 marks]
4. Solve the following
 - (a) $\log_2 2x = 2\log_4 (x + 4)$ [6 marks]
 - (b) $2\left|\frac{x-3}{2x-1}\right| \geq 1$ [7 marks]
5. Given a function $f(x) = \ln(2x + 1)$
 - (a) State the domain and range of $f(x)$ [2 marks]
 - (b) Find the inverse function of $f(x)$ and state its domain and range. Hence, find the value of x for which $f^{-1}(x) = 0$ [7 marks]
 - (c) Sketch the graph of $f(x)$ and $f^{-1}(x)$ on the same coordinate axes. [3 marks]

6. (a) A function $f(x)$ is defined as

$$f(x) = \begin{cases} \frac{9-x^2}{x-3} & x < 3 \\ -6 & x = 3 \\ \frac{162x-54x^2}{x^3-27} & x > 3 \end{cases}$$

Determine whether $f(x)$ is continuous at $x = 3$. [7 marks]

- (b) Find $\lim_{x \rightarrow \infty} \frac{5x^3 + 4x - 9}{7 - 3x^3}$ [3 marks]

7. (a) Given $y = x^2 e^{-3x}$. Find $\frac{d^2 y}{dx^2}$. Give your answer in the simplest form. [5 marks]

- (b) Given $x^2 y = \sin(2x^2 + \pi)$. Show that

$$2y(1 + 8x^4) + 4x \frac{dy}{dx} + x^2 \frac{d^2 y}{dx^2} = 4 \cos(2x^2 + \pi) \quad [7 \text{ marks}]$$

8. Given $f(x) = x^2 - \frac{6}{x}$ where $x \neq 0$. Find the coordinate(s) of stationary point and determine whether it is a relative maximum or relative minimum. Give the coordinates correct to 3 decimal places. [5 marks]

9. The function $g(x)$ is defined by

$$g(x) = \begin{cases} m^2 - 8 & x \leq 2 \\ \frac{x-2}{\sqrt{2x-2}} & 2 < x \leq 8 \\ \frac{|8-x|}{x-8} + k & x > 8 \end{cases}$$

Where m and k is constant.

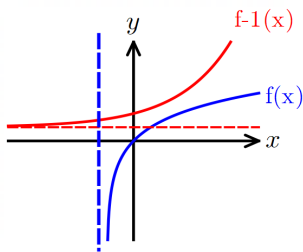
- (a) If $\lim_{x \rightarrow 2} g(x)$ exists, find the values of m . [5 marks]

- (b) Find the values of k such that $g(x)$ is discontinuous at $x = 8$. [3 marks]

10. Given a right circular cylinder with height $2h$ and radius $r = \sqrt{a^2 - h^2}$, which is inscribed in a sphere of fixed radius a .
- (a) Show that the volume of the cylinder is $V = 2\pi(a^2 - h^2)h$. [5 marks]
- (b) From part (a), find the maximum value of the volume, V in terms of π as the height, h varies. Hence, state the value of the volume if $a = 3$. [9 marks]
- (c) Show that the ratio of the volume of the sphere to the maximum volume of the cylinder is $3\sqrt{3} : 1$. [3 marks]

FINAL ANSWER

- Q1 (a) $z_1^2 = -1 + 0i$ $\overline{z_2} = 2 - i\sqrt{3}$ (b) $|w| = 2$, $\arg w = \frac{\pi}{6}$
- Q2 (a) $x = 1$ (b) $(-4, -2] \cup [2, 3)$
- Q3 (a) $f(x) = \frac{x+1}{2x+1}$ $f\left(\frac{1}{2}\right) = \frac{3}{4}$ (b) $x \neq 0, x \neq -1, x \neq -\frac{1}{2}$
- Q4 (a) $x = 4$ (b) $\left(-\infty, \frac{1}{2}\right) \cup \left(\frac{1}{2}, \frac{7}{4}\right]$
- Q5 (a) $D_f = \left(-\frac{1}{2}, \infty\right)$ $R_f = (-\infty, \infty)$ (b) $f^{-1}(x) = \frac{e^x - 1}{2}$ $D_{f^{-1}} = (-\infty, \infty)$
 $, R_{f^{-1}} = \left(-\frac{1}{2}, \infty\right), x = 0$
- Q6 (a) $f(x)$ is continuous at $x = 3$ (b) $-\frac{5}{3}$
- Q7 (a) $\frac{d^2y}{dx^2} = e^{-3x}[9x^2 - 12x + 2]$ (b) DIY
- Q8 $(-1.44, 6.241)$ is a relative minimum
- Q9 (a) $m = \pm\sqrt{10}$ (b) $k = \mathbb{R} / \{2\}$
- 10 (a) DIY (b) $V = 2\pi a^2 h - 2\pi h^3$,
(c) DIY



2020/21

SECTION A

1. (a) Given the function

$$f(x) = \begin{cases} x+1 & x < 1 \\ \frac{x^2 - x - a}{x - a} & x \geq 1, x \neq a \end{cases}$$

Is continuous at $x = 1$. Find the values of a . Hence, determine the $\lim_{x \rightarrow a} f(x)$.

[6 marks]

(b) Evaluate $\lim_{x \rightarrow +\infty} \left(\frac{2x+5}{3x-4} \right)^{\frac{1}{3}}$.

[3 marks]

2. (a) Given the parametric equation

$$x = \frac{1}{t} + 3t^2, y = t^2 - 4\sqrt{t}, t > 0$$

Find $\frac{d^2y}{dx^2}$ in terms of t . Hence, evaluate $\frac{d^2y}{dx^2}$ at $t = 1$.

[6 marks]

(b) Given $y = \ln \left(\frac{\sin 2x}{1 - \cos 2x} \right)$. Show that $\frac{dy}{dx} = -2 \operatorname{cosec} 2x$.

[5 marks]

3. A farmer has a 100 meter of fencing wants to fence off a rectangular piece of land that borders a straight river. The part of the land which borders the river needs no fencing. Find the dimensions of the enclosed region that has the maximum area.

[5 marks]

SECTION B

1. (a) Solve the equation $\ln 2^{4x-1} = \ln 8^{x+5} + \log_2 16^{1-2x}$ by expressing your answer in terms of $\ln 2$.

[4 marks]

(b) Given $z_1 = 2 - 3i$ and $z_2 = 4 + 2i$. Without using calculator, find $z_3 = \frac{\overline{z_1}}{z_2}$ and

hence, evaluate $|z_3|$.

[5 marks]

2. Determine the solutions of the following inequalities

(a) $\frac{4x}{4-x} \geq 6$. [5 marks]

(b) $|x-6| < 3|x-2|$. [6 marks]

- 3.

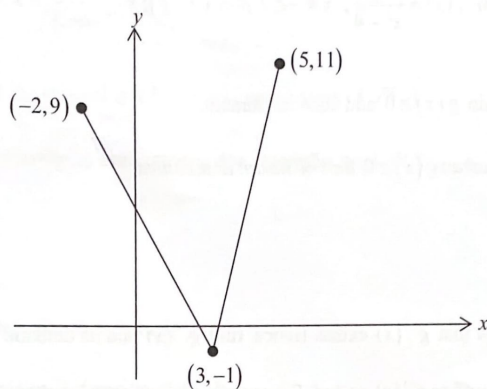


FIGURE 3
RAJAH 3

Figure 3 shows a function $f(x)$ with domain $-2 \leq x \leq 5$

- (a) State the range of $f(x)$. [1 mark]
 (b) Determine the function of $f(x)$. [4 marks]
 (c) Evaluate $(f \circ f)(0)$. [2 marks]

4. Given a function $f(x) = \frac{3}{x^2-4}, x \neq -2, 2$ and $(f \circ g)(x) = \frac{3}{x-5}, x \neq 5$

- (a) Obtain $g(x) \geq 0$ and state its domain. [4 marks]
 (b) Show that $g^{-1}(x)$ exists. Hence, find $g^{-1}(x)$ and its domain. [5 marks]
 (c) Find the values of a and b if $h(x) = ax + b$, such that $(h \circ g^{-1})(x) = 5x^2 + 12$. [4 marks]

5. Given $f(x) = \frac{2x^2 + 2}{x^2 - 4}$.
- (a) Find the vertical and horizontal asymptotes of $f(x)$. [8 marks]
- (b) Hence, sketch the graph $f(x)$. [3 marks]
6. (a) Given $y = \tan\left(\frac{\pi}{4} + x\right)$, show that $\frac{d^2y}{dx^2} - 2y \frac{dy}{dx} = 0$. [5 marks]
- (c) Differentiate $g(x) = \frac{e^x}{e^x - 2}$, $x \neq \ln 2$ with respect to x . [6 marks]
- (d) Find the value of tangent to the curve $y^2 = x^3$ and $2x^2 + 3y^2 = k$ where k is a constant, at the point $(1, -1)$ and $(1, 1)$. [5 marks]
7. Consider a cone as shows in figure 7.

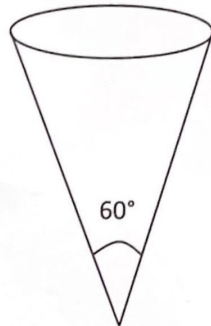


FIGURE 7
RAJAH 7

Water flows out from a cone through a hole located at the bottom tip at the rate of $5\text{cm}^3 / \text{s}$. Find the rate of change of the height of the water when the volume of water in the cone is $3000\pi\text{cm}^3$. [8 Marks]

FINAL ANSWER**SECTION A**

$$1. \quad (a) a = 2, \lim_{x \rightarrow 2} = 3 \quad (b) \sqrt[3]{\frac{2}{3}} / 0.8736$$

$$2. \quad (a) \left. \frac{d^2 y}{dx^2} \right|_{t=1} = \frac{3}{25} / 0.12 \quad (b) DIY$$

$$3. \quad x = 50, y = 25$$

SECTION B

$$1. \quad (a) x = \frac{4 + 16 \ln 2}{8 + \ln 2} \quad (b) z_3 = \frac{7}{10} + \frac{2}{5}i, |z_3| = 0.8062$$

$$2. \quad (a) \left\{ x : \frac{12}{5} \leq x < 4 \right\} \quad (b) \{ x : x < 0 \text{ or } x > 3 \}$$

$$3. \quad (a) R_f = [-1, 11] \quad (b) f(x) = \begin{cases} -2x + 5 & , -2 \leq x \leq 3 \\ 6x + 9 & , 3 \leq x \leq 5 \end{cases} \quad (c) f \circ f(0) = 11$$

$$4. \quad (a) g(x) = \sqrt{x-1}, D_g = [1, \infty) \quad (b) \text{show one to one } g^{-1}(x) = x^2 + 1, D_{g^{-1}} = [0, \infty) \\ (c) a = 5, b = 7$$

$$5. \quad (a) VA : x = -2, x = 2 \quad HA : y = 2 \quad (b) DIY$$

$$6. \quad (a) DIY \quad (b) x = 0 \\ (c) \text{Value of tangent}$$

$$\text{at } (1, -1) \Rightarrow \frac{dy}{dx} = \frac{2}{3}$$

$$\text{at } (1, 1) \Rightarrow \frac{dy}{dx} = -\frac{2}{3}$$

$$7. \quad \frac{dh}{dt} = -\frac{1}{60\pi}$$

2022/23

SECTION A

1. Solve

(a) $\lim_{x \rightarrow -3} \frac{x^3 + 27}{x + 3}$ [4 marks]

(b) $\lim_{x \rightarrow 2} \frac{\sqrt{x^2 + 12} - 4}{x - 2}$ [5 marks]

2. By expressing $y = \frac{1}{(x-1)^2(x-3)}$ into partial fractions, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. Give your

answer in the simplest form. Hence, evaluate $\frac{d^2y}{dx^2}$ when $x = -1$. [9 marks]

3. Find the critical numbers for $f(x) = x^4 - 8x^3 + 18x^2 - 27$. Hence, determine the points of inflection. [7 marks]

SECTION B1. Given $z_1 = 6 + 3i$ and $z_2 = 5 - 2i$

(a) Verify $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$. [3 marks]

(b) Find $\arg(z_1)$. [2 marks]

2. Solve

(a) $x^{\ln x - 2} = e^3$. [7 marks]

(b) $|2x + 3| > |x - 5|, x \neq 5$ [4 marks]

3. Given $f(x) = 4^{x+2}$

(a) Find the value of a when $f(a) = 1$. [2 marks]

(b) Find f^{-1} and its domain [4 marks]

4. Given $f(x) = e^{-2x} + 3; (g \circ f)(x) = \frac{1}{2} \ln(e^{-2x} + 2)$

(a) Find domain and range of $f(x)$. [3 marks]

(b) Find $g(x)$ [5 marks]

(c) Hence, find $g^{-1}(x)$. Sketch the graph of $g(x)$ and $g^{-1}(x)$ on the same axes. [6 marks]

5. (a) Evaluate $\lim_{x \rightarrow -\infty} \frac{3x + 2}{\sqrt{4x^2 - 3}}$ [4 marks]

(b) Given $f(x) = \begin{cases} ax + 3 & x > 5 \\ 8 & x = 5 \\ x^2 + bx - 2a & x < 5 \end{cases}$

Find a and b such that the function is continuous for all x . [4 marks]

(c) Given $y = \frac{x + 1}{(x - 4)(x + 2)}$

By finding the limit at appropriate points, obtain the horizontal and vertical asymptotes. [6 marks]

6. (a) Given $y^3 = e^{x-y}$.

Show that

(i) $\frac{dy}{dx} = \frac{y}{3 + y}$ [4 marks]

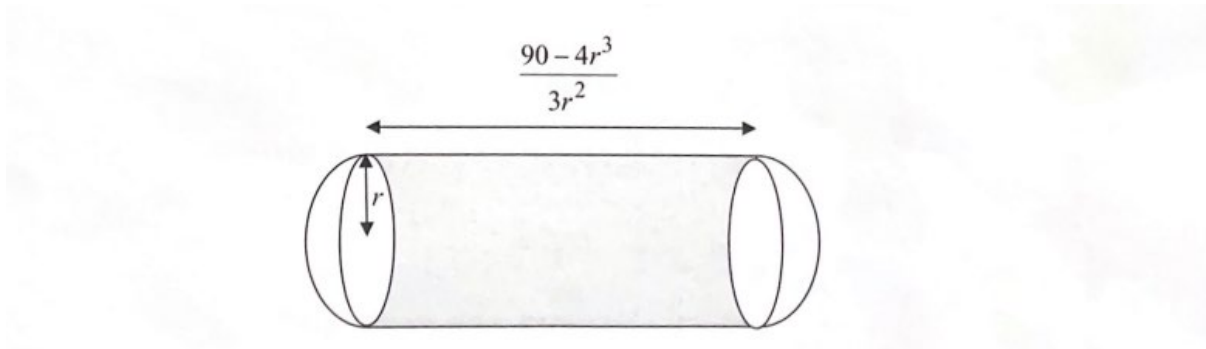
(ii) $(3 + y)^3 \frac{d^2y}{dx^2} - 3y = 0$ [4 marks]

(b) Given the parametric equations for the curve

$$x = 2 + \sqrt{\cos 2t} \quad y = 1 - \sqrt{\sin 2t}$$

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. Give your answer in the simplest form. [8 marks]

7. The figure below shows a cylinder with both ends are two hemispheres with radius r meter and length $\frac{90 - 4r^3}{3r^2}$ meter.



- (a) Show that the total surface area is $A = \frac{4}{3}\pi r^2 + \frac{60\pi}{r}$ [3 marks]
- (b) Hence, find the value of r when the total surface area is minimum. [6 marks]

FINAL ANSWER

SECTION A

- (a) 27 (b) $\frac{1}{2}$
- $$y = -\frac{1}{4(x-1)} - \frac{1}{2(x-1)^2} + \frac{1}{4(x-3)}$$

$$\frac{dy}{dx} = \frac{1}{4(x-1)^2} + \frac{1}{(x-1)^3} - \frac{1}{4(x-3)^2}$$

$$\frac{d^2y}{dx^2} = -\frac{1}{2(x-1)^3} + \frac{3}{(x-1)^4} + \frac{1}{2(x-3)^3}$$

$$\left. \frac{d^2y}{dx^2} \right|_{x=-1} = -\frac{17}{128}$$
- Critical number $x = 0, x = 3$, point of inflections $(3, 0)$

SECTION B

1. (a) DIY (b) $Arg(z_1) = 0.4363rad$
2. (a) $x = e^3, x = \frac{1}{e}$ (b) $(-\infty, -8) \cup \left(\frac{2}{3}, 5\right) \cup (5, \infty)$
3. (a) $a = -2$ (b) $f^{-1}(x) = \frac{\ln x}{\ln 4} - 2, D_{f^{-1}} = (0, \infty)$
4. (a) $D_f = (-\infty, \infty) R_f = (3, \infty)$
 (b) $g(x) = \frac{1}{2}\ln(x-1)$
 (c) $g^{-1}(x) = e^{2x} + 1$
5. (a) $-\frac{3}{2}$ (b) $a = 1, b = -3$ (c) $VA : x = 4, x = -2$
 $HA : y = 0$
6. (a)i,ii DIY (b) $\frac{dy}{dx} = \left(\frac{\cos 2t}{\sin 2t}\right)^{\frac{3}{2}} = (\cot 2t)^{\frac{3}{2}}, \frac{d^2y}{dx^2} = \frac{3 \cos 2t}{(\sin 2t)^{\frac{7}{2}}}$

$$\frac{d^2A}{dr^2} = \frac{120\pi}{r^3} + \frac{8}{3}\pi$$

$$\left. \frac{d^2A}{dr^2} \right|_{r=2.82} = 25.133 > 0 (\text{min})$$

- 7.(a)DIY (b) $\therefore r = 2.82$ will give minimum surface area.

2023/24

SECTION A

1. Find

(a) $\lim_{x \rightarrow 2} \frac{10 - 3x - x^2}{x - 2}$ [3 marks]

(b) $\lim_{x \rightarrow \infty} \frac{x + 2}{x - 1}$ [3 marks]

(c) $\lim_{x \rightarrow 3^-} \frac{|x - 3|}{x^2 - 2x - 3}$ [4 marks]

2. (a) Given $f(x) = \begin{cases} x^2 - 3 & x < 2 \\ \sqrt{x - 1} & x \geq 2 \end{cases}$

Determine whether $f(x)$ is differentiable at $x = 2$. [5 marks]

(b) Differentiate $y = (x^2 + 1)(\sqrt{1 - x^2})$ with respect to x . Give your answer in the Simplest form. [4 marks]

3. A liquid is to be cleared of sediment using a right circular shape cone filter. Assume that the height of the cone is 16cm and the radius at the base of the cone is 4cm . Given that the volume of the liquid is decreasing at a rate of 2cm^3 per minute, find the rate of change in the liquid level when the depth of the liquid is 8cm . [6 marks]

SECTION B

1. Given that $-2m + (3 + i)n = -4 + 3i$. Find the complex numbers m and n if n is the conjugate of m . [5 marks]

2. (a) Given $\log_9 \left(\log_{\sqrt{p}} \frac{1}{27} \right) = \frac{1}{2}$. Find the value of p . [4 marks]

(b) Solve $2^x + 4 < 2^{2x} + 2^x < 12$ [8 marks]

3. (a) Given $f(x) = \frac{2e^{\frac{3}{2}x}}{e^{3x} - 1}$ and $g(x) = \frac{e^{3x} + 1}{e^{3x} - 1}$. Evaluate $[f(x)]^2 - [g(x)]^2$ [4 marks]
- (b) Given $g(x) = (x - 2)^2 - 6$ and $h(x) = 3x^2 - 12x + 2$. Show that $f(x) = 3x + 8$ if $(f \circ g)(x) = h(x)$. [4 marks]
4. Given $f(x) = 2 - \sqrt{3 - x}$
- (a) Find the domain and range of $f(x)$. [2 marks]
- (b) Show that $f(x)$ is a one-to-one function by using algebraic approach. [2 marks]
- (c) Determine $f^{-1}(x)$ and its domain [4 marks]
- (d) Hence, sketch the graph of $f(x)$ and $f^{-1}(x)$ on the same axes. Label your graph completely. [3 marks]
5. (a) Find the value of a if $\lim_{x \rightarrow a} \frac{\sqrt{x} - 2}{x - 4} = \frac{1}{6}$ [3 marks]
- (b) The function $f(x)$ is given by $f(x) = \begin{cases} 1 - 2e^{2x} & x < 0 \\ 1 & x = 0 \\ e^{3x} - 2 & x > 0 \end{cases}$
- (i) Determine the existence of limit of $f(x)$ as x approaches 0. [3 marks]
- (ii) Determine whether $f(x)$ is continuous at $x = 0$. Hence, state the interval(s) where $f(x)$ is continuous. [3 marks]
- (c) Given $f(x) = \frac{\sqrt{x^3 + 5}}{x - 2}$. By finding the limit appropriate values, obtain the horizontal and vertical asymptotes, if they exist. [5 marks]

6. (a) Given $xy - \sin(xy) - 3 = 0$

(i) Find $\frac{dy}{dx}$ in terms of x and y . [4 marks]

(ii) Hence, show that $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0$ [2 marks]

(b) Given the parametric equations $x(t) = \ln(t^2 + t)$ and $y(t) = \sqrt{t^2 + t}$ for $t > 0$.

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of t . Hence, evaluate $\frac{d^2y}{dx^2}$ when $x = \ln 2$. [10 marks]

7. Given $f(x) = \cos x + \sin x, 0 \leq x \leq 2\pi$

(a) By using the first derivative test, find the extremum points. [5 marks]

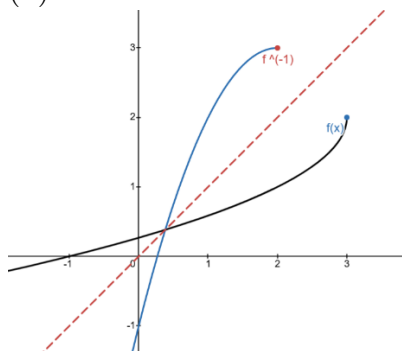
(b) Determine the inflection points of $f(x)$ [4 marks]

FINAL ANSWER**SECTION A**

1. (a)-7, (b)1 (c) $-\frac{1}{4}$
2. (a)f is not differentiable at $x = 2$ (b) $\frac{dy}{dx} = \frac{-3x^3 + x}{\sqrt{1-x^2}}$
3. $\frac{dh}{dt} = -\frac{1}{2\pi} / -0.159$

SECTION B

1. $m = -\frac{17}{6} - \frac{7}{6}i, n = -\frac{17}{6} + \frac{7}{6}i$
2. (a) $p = \frac{1}{9}$ (b) $\{x : 1 < x < \log_2 3\}$
3. (a) -1 (b) $f(x) = 3x + 8$
4. (a) $D_{f^-} = (-\infty, 3], R_f = (-\infty, 2]$ (b) DIY
(c) $f^{-1}(x) = 3 - (2 - x)^2 / -x^2 + 4x - 1, D_{f^{-1}} = (-\infty, 2]$
(d)



5. (a) $a=16$ (b)(i) limit exists
(ii) f is not continuous at $x=0$, f is continuous on $(-\infty, 0) \cup (0, \infty)$
(c) VA : $x = 2$, HA d.n.e
6. (a)i. $\frac{dy}{dx} = \frac{y \cos xy - y}{x - x \cos xy} / \frac{y(\cos xy - 1)}{x(1 - \cos xy)} / \frac{-y(1 - \cos xy)}{x(1 - \cos xy)} = \frac{-y}{x}$ ii. DIY
(b) $\frac{dy}{dx} = \frac{1}{2} \sqrt{t^2 + t}, \frac{d^2y}{dx^2} = \frac{1}{4} \sqrt{t^2 + t}, \frac{d^2y}{dx^2} \Big|_{t=1} = \frac{\sqrt{2}}{4}$
7. (a) $\max = \left(\frac{\pi}{4}, \sqrt{2}\right), \min = \left(\frac{5\pi}{4}, -\sqrt{2}\right)$ (b) inflection points : $\left(\frac{3\pi}{4}, 0\right), \left(\frac{7\pi}{4}, 0\right)$

2024/25

SECTION A

1. Evaluate the following limits

(a) $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 9}$ [5 marks]

(b) $\lim_{x \rightarrow 2} \frac{x - 2}{\sqrt{3x - 4} - \sqrt{x}}$ [4 marks]

2. (a) Find the derivative of $f(x) = \frac{3}{x^2 - 2}$ using the first principle. [5 marks]

(b) Find $\frac{dy}{dx}$ for $y = \ln(x^2 \sqrt{1 - x^2})$ [4 marks]

3. Water flows into an upright cone of radius 5cm and height 8cm at the rate of $2\text{cm}^3 / \text{s}$ as shown in **FIGURE 3**, with h is water level and r is radius of water surface.

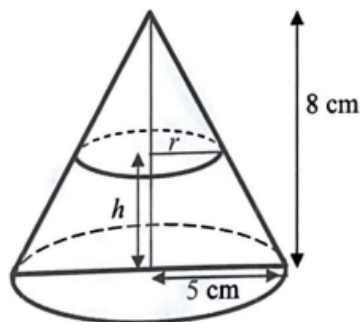


FIGURE 3

Find the rate of decreasing radius of water surface when radius is 2cm [7 marks]

SECTION B

1. Without using calculator, express $\frac{4-2i}{-i^5(2-3i)}$ in the form of $a+bi$. Hence, find the exact value of $\left| \frac{4-2i}{-i^5(2-3i)} \right|$. [5 marks]
2. Solve
- (a) $\log_2 x + 2\log_4(10-x) = 4$ [5 marks]
- (b) $\frac{2x^2 + 5x - 3}{x^2 - 5x + 6} < 2$ [6 marks]
3. Functions g and $f \circ g$ are defined by $g(x) = e^{x-5}$ and $f \circ g(x) = \sqrt{5x}$ for all $x \geq 0$
- (a) Find the function f and determine its domain. [5 marks]
- (b) Determine the value of $(g \circ f)(e^2)$. [2 marks]
4. Given $f(x) = \frac{1}{x-3}, x > 3$
- (a) Find all x such that $f(x) \geq 3$ [2 marks]
- (b) (i) Show algebraically that $f(x)$ is a one-to-one function. [3 marks]
- (ii) Determine the inverse function of $f(x)$. [3 marks]
- (iii) State the domain and range for $f^{-1}(x)$. [2 marks]
- (iv) Sketch the graph of $f(x)$ and $f^{-1}(x)$ on the same axes. [3 marks]
5. Consider
- $$w(x) = \begin{cases} x^2 - 4x + 3k + 10 & x \leq 0 \\ 2e^x - 5k & 0 < x < 3 \\ \frac{2x-1}{x+2} & x > 3 \end{cases}$$
- (a) Find $\lim_{x \rightarrow 0^+} w(x)$ and $\lim_{x \rightarrow 0^-} w(x)$. Hence, determine the value of k such that the function $w(x)$ is continuous at $x = 0$. [5 marks]

- (b) Determine whether $w(x)$ is continuous at $x = 3$. State your reason. [2 marks]
- (c) Determine the horizontal asymptote for $x > 3$. [3 marks]
- (d) Sketch the graph for $w(x)$. Hence, find the range of the function. [4 marks]

6. (a) Given $x^2 \cos y = \sin y^3$, find $\frac{dy}{dx}$. [4 marks]

- (b) The parametric equations of a curve are given by

$$x = e^t + 1 \text{ and } y = e^t + 9e^{-t}$$

- (i) Find the coordinate of the stationary point. [6 marks]

- (ii) Find $\frac{d^2y}{dx^2}$ in terms of x only. Hence, show that

$$(x - 1)^3 \frac{d^2y}{dx^2} - 18 = 0 \quad [6 \text{ marks}]$$

7. **FIGURE 7** shows a semicircle with diameter $2x$ units attached to a rectangle with length $2x$ units and width y unit. The perimeter of the figure is 60 units. Show that the maximum area is $\frac{1800}{4 + \pi} \text{ unit}^2$.

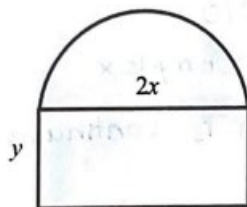


FIGURE 7
RAJAH 7

[9 marks]

FINAL ANSWER**SECTION A**

1. (a) $\frac{9}{2}$ (b) $\sqrt{2}$

2. (a) $f'(x) = -\frac{6x}{(x^2 - 2)^2}$ (b) $\frac{dy}{dx} = \frac{2}{x} + \frac{1}{2} \left(\frac{-2x}{1 - x^2} \right)$

3. $\frac{dr}{dt} = -\frac{5}{16\pi} / -0.0995 \text{ cm s}^{-1}$

SECTION B

1. $-\frac{8}{13} + \frac{14}{13}i, \frac{2\sqrt{65}}{13}$

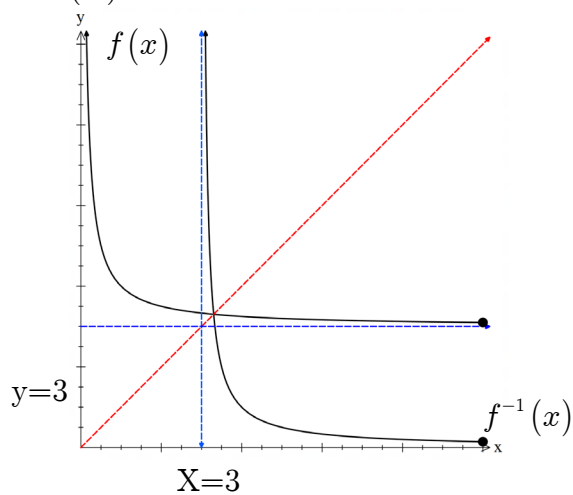
2. (a) $x = 2, 8$ (b) $(-\infty, 1) \cup (2, 3)$

3. (a) $f(x) = \sqrt{25 + 5 \ln x}, D_f = [e^{-5}, \infty)$ (b) $2.50/2.449$

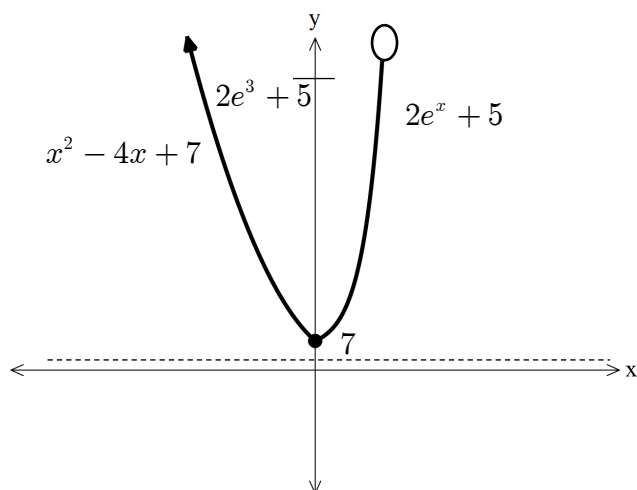
4. (a) $\left\{ x : 3 < x \leq \frac{10}{3} \right\}$ (b)(i) DIY (ii) $f^{-1}(x) = \frac{1}{x} + 3$

(iii) $D_{f^{-1}} = R_f = (0, \infty)$ $R_{f^{-1}} = D_f = (3, \infty)$

(iv)



5. (a) $k = -1$ (b) not continuous (c) $y = 2$ is the horizontal asymptote
 (d) $\text{Range} = (1, 2) \cup [7, \infty)$



6.(a) $\frac{dy}{dx} = \frac{2x \cos y}{x^2 \sin y + 3y^2 \cos y^3},$

(b) (i) stationary points = $(4, 6)$

(ii) $\frac{d^2y}{dx^2} = \frac{18}{(x-1)^3}, \text{DIY}$

7. $\frac{dA}{dx} = 60 - (4 + \pi)x$ $\frac{d^2A}{dx^2} = -(4 + \pi) < 0 [\text{max}]$